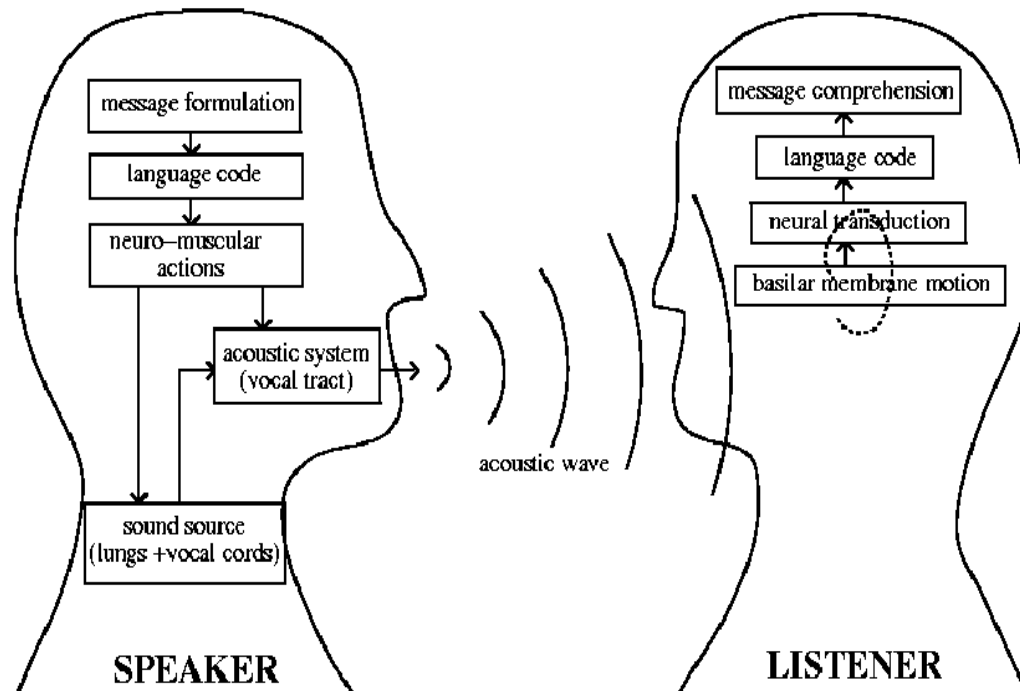


Forensic Audio

TEJASVI
BHATIA

Speaker & Listener

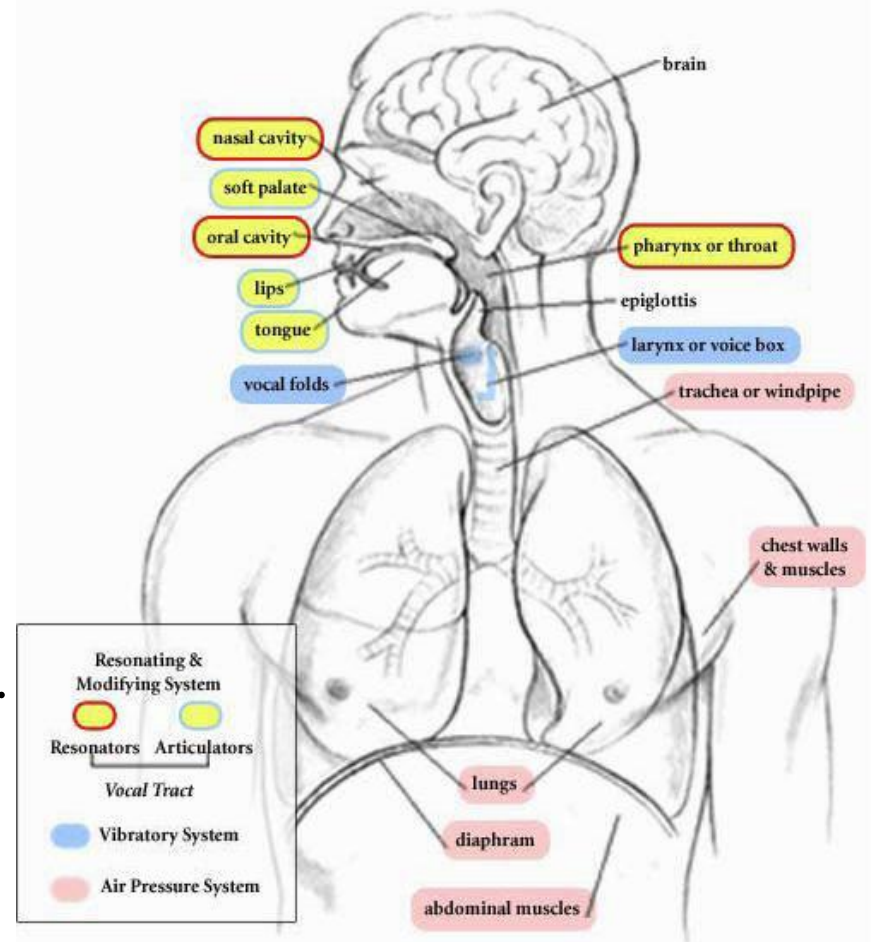


Production of vibrational energy by articulation after brain instructs to perform.

Eardrums convert this vibrational energy into signals that travel along nerves to the brain, which interprets them as voices, music, noise, etc.

Production of speech

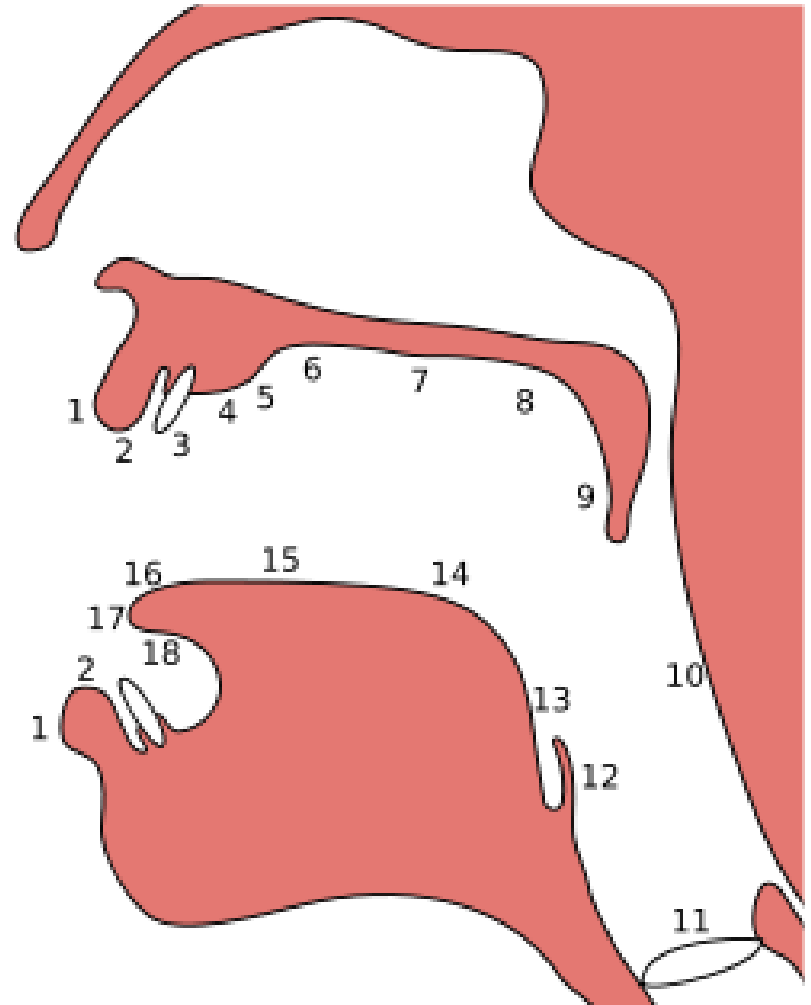
- The basic responsible function for production of speech are: Generation of air pressure, regulation of vibration, and control of resonators
- Larynx sometimes called voice box is the most important organ among Lung, Vocal code, *pharynx*, Tongue, Teeth & Lip etc.
- Yes Tongue is also the valuable articulatory organ



Place of articulation

Places of articulation :

- 1. Exo-labial,
- 2. Endo-labial,
- 3. Dental,
- 4. Alveolar,
- 5. Post-alveolar,
- 6. Pre-palatal,
- 7. Palatal,
- 8. Velar,
- 9. Uvular,
- 10. Pharyngeal,
- 11. Glottal,
- 12. Epiglottal,
- 13. Radical,
- 14. Postero-dorsal,
- 15. Antero-dorsal,
- 16. Laminal,
- 17. Apical,
- 18. Sub-apical



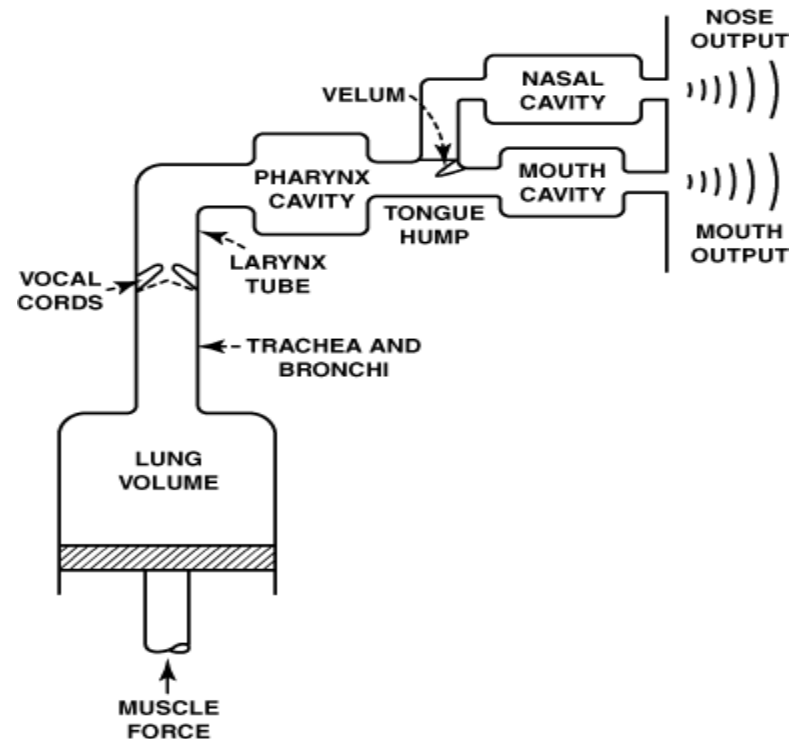
Why animal can't produce speech

- Except Human being, any other animals can't produce speech because they don't have co-articulation system

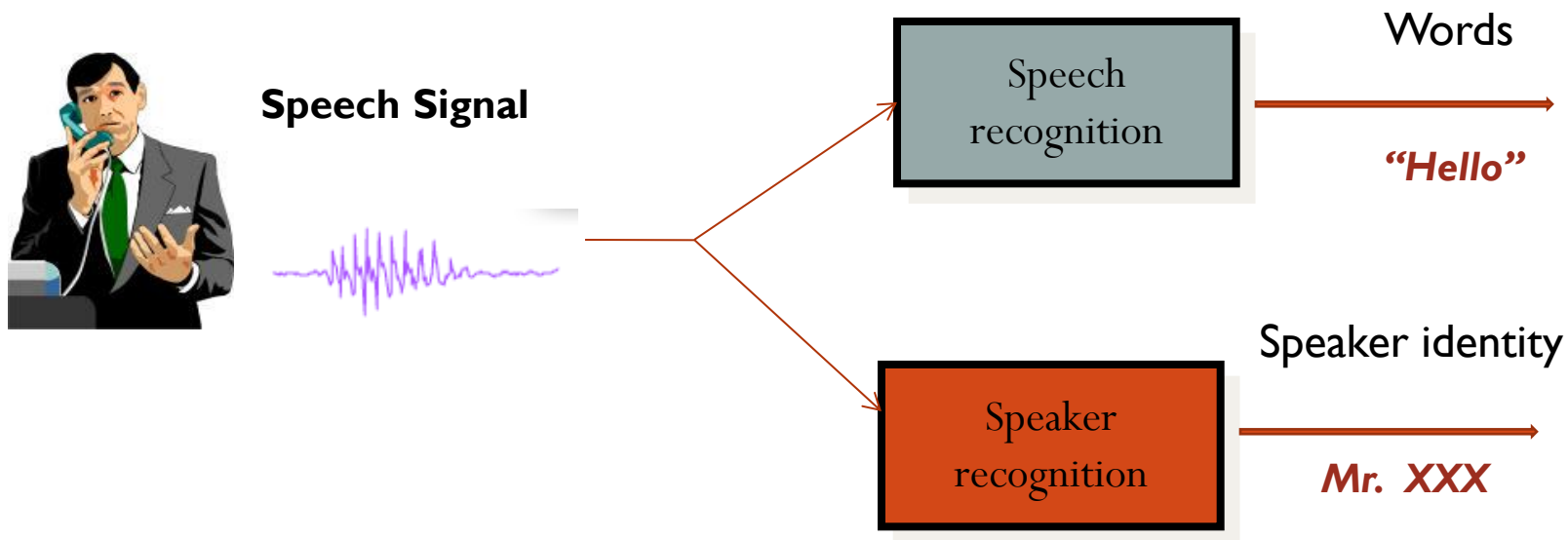
SOUND IS NOT CONTROLLED

Why person to person different in voice

- As vocal tracks are different in shape, size and length and the tension of the vocal folds, the voice production is also different person to person



SPEAKER IDENTIFICATION



Speaker Recognition

- The process of automatically recognizing who is speaking on the basis of individual's speech signals information.
- It is divided into two categories:
 - Speaker Identification
 - The task of determining an unknown speaker's identity. (Speaker identification determines which registered speaker provides a given utterance from amongst a set of known speakers.)
 - Speaker Verification
 - With certain identity the voice is used to verify (Speaker verification accepts or rejects the identity claim of a speaker - is the speaker the person they say they are?)
- In forensic applications, it is suggested to first perform a speaker identification process to create a list of "best matches" and then perform a series of verification processes to determine a conclusive match

When Speaker Identification is required

- After criminal offence, no evidence was left except the recorded voice/voices of the suspect/suspects then we definitely need for Speaker Identification.
- YES, THERE IS LIMITATION ALSO:
 - If the signal is distorted and not having much information..

Why Speaker Identification

- Speaker Identification is carried out by a combination of auditory and acoustic methods (and, where appropriate, some text analysis methods) and provides an opinion as to whether a particular voice, for example recorded making a telephone call, or participating in a conversation recorded by a recording device, is that of a particular known person.
- Speaker Identification is essential for several criminal offences, such as making hoax calls to the police, ambulance or fire brigade, making threatening or harassing telephone calls, blackmail or extortion demands, or taking part in criminal conspiracies such as those involving the importation, trafficking or manufacture of illegal drugs etc.

POSSIBILITIES IN SPEAKER IDENTIFICATION

- Determine the speaker identity
- Selection between a set of known voices
- The user does not claim an identity
- **Closed set** identification
 - Assume that all speakers are known to the system
- **Open set** identification
 - Possibility that speaker is not among the speakers known to the system

Closed set & Open set identification

- "Closed-set identification": The task of identifying an unidentified speaker within a known database.
- "Open-set identification": The task of identifying a known speaker within the unknown database.
- In Forensic Speech Analysis, the cost-set-identification is common one.

Common Facing Forensic Audio Cases

- Kidnapping for ransom
- Anonymous calls, threatening calls
- Obscene calls
- Drug peddling
- Sharing of vital information across the border (say. anti-social person)
- Bribery
- Match fixing
- And many more ...

Problems in Forensic speaker examination

- Recorded samples
 - Noisy (SNR 5-6dB or less)
 - Distorted/damped & short duration
 - Non-contemporary
 - Disguised
 - Different texts
- Mode of Recording
 - Telephone
 - Cellular phone
 - Tape recorder etc

Problem in Forensic speaker examination contd..

- Speakers/suspects may suffer from a number intrusive conditions
 - Effects of health (cold, fever)
 - Ingested drugs
 - Emotional states(fear or stress)
- Speakers/suspects - non co-operative etc.

Essential parameters of good quality audio

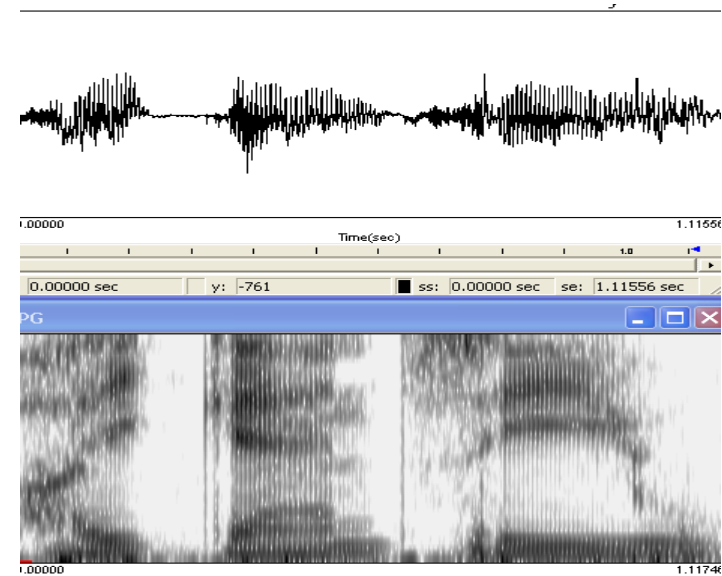
- If the signal contain maximum information without any noise then we can say the signal has good quality sample.
- It is possible only when recording in noise proof area using sophisticated device by a professional person.
- EXACTLY, NOISE ARE THE ENEMY OF THE SPEECH SAMPLE!

Methods of speaker identification

- Aural examination of voice
- Spectrographic - visual examination of voice spectrogram
- Semiautomatic/ Automatic method of voice identification

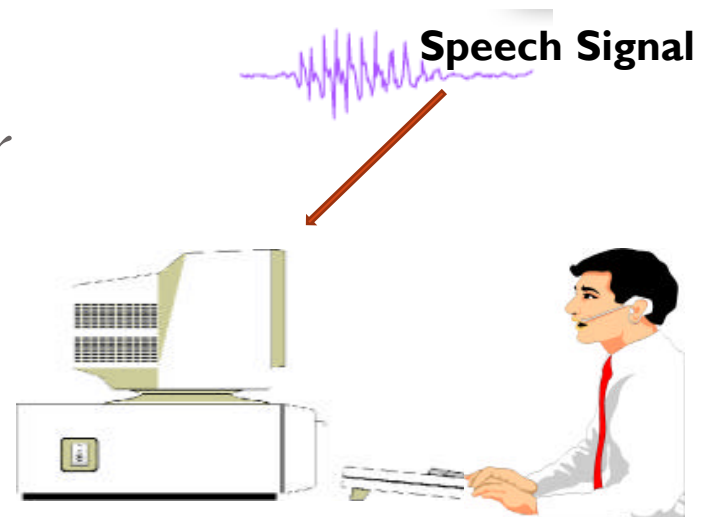
METHODS FOR SPEAKER IDENTIFICATION

- Auditory /Aural examination method
- Spectrographically visual-analysis method viz.
 - *Computerized Speech Laboratory (CSL)*
 - *Multi-Speech*



...Continued

- Automatic Speaker identification Systems viz.
 - *Text Independent Speaker Identification System (SPID)*
 - *Language Independent Speaker Identification System (LISIS)*

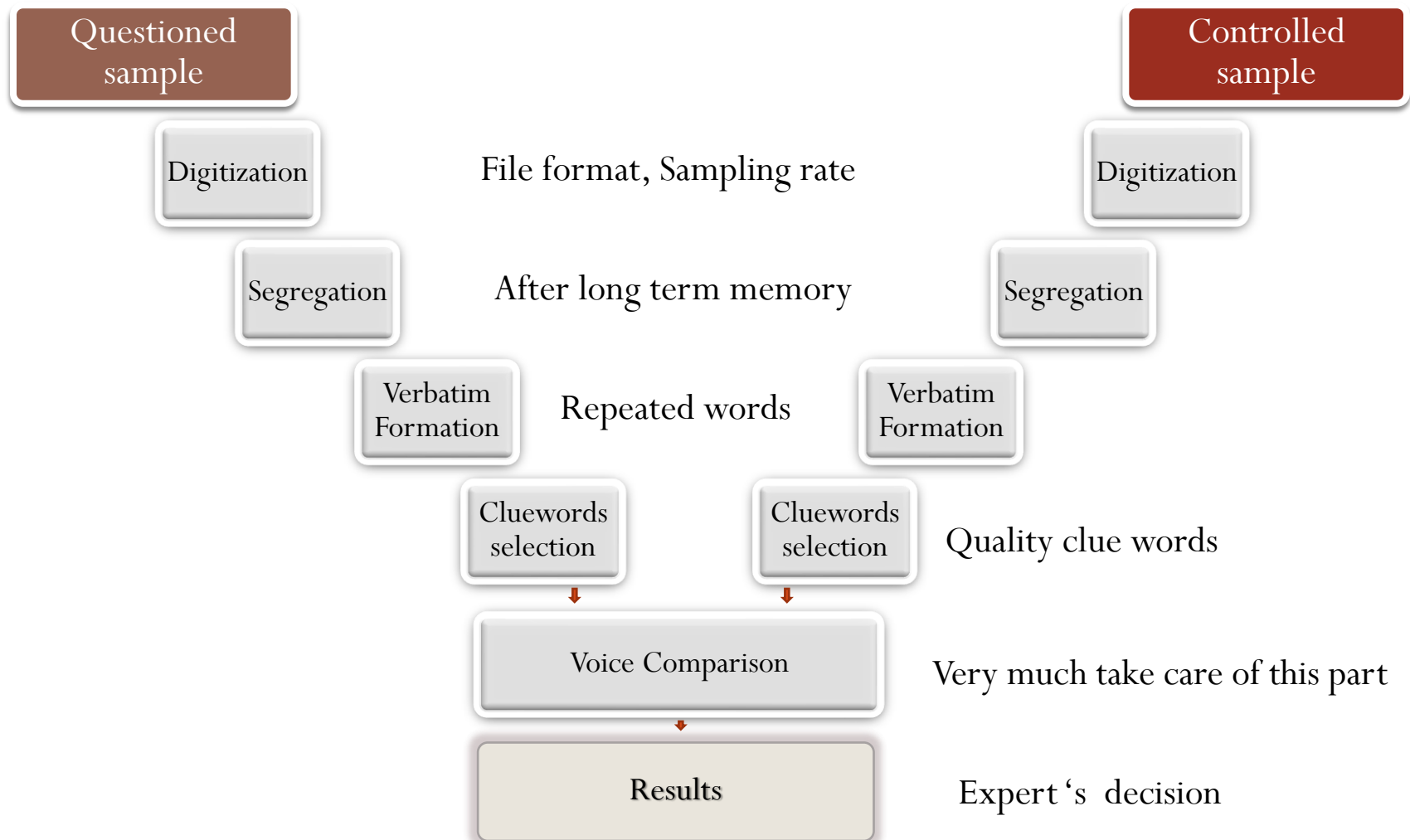


Audio Examination Tools

- Audio High-End Professional System
- Digitization Tools
 - Professional Audio System
- Pre analysis tools
 - Goldwave, Adobe Audition, Cool Edit etc.
- Analysis Software
 - Computerized speech Lab, Multispeech etc.
 - Semi Automatic SPID

Process of Speaker Identification

Each steps will not compromise any mistake



Thank you for your kind presence